

RANDOLPH AFB, TX, USA

WMO: 722536

Lat: 29.533N

Lon: 98.262W

Elev: 728

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.6	32.0	11.9	10.4	41.0	16.4	12.9	45.3	26.2	48.8	24.1	52.9	9.0	330	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	100.1	73.8	98.6	73.8	96.6	74.0	78.5	87.1	78.0	86.7	77.3	86.1	10.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	145.3	80.0	76.3	140.9	79.8	75.4	136.5	79.6	42.7	87.2	41.9	87.0	41.2	85.9	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.5	17.7	DB	22.1	103.6	4.0	3.5	19.2	106.1	16.9	108.1	14.7	110.1	11.8	112.6	
			WB	19.7	80.4	3.2	1.2	17.4	81.2	15.6	81.9	13.8	82.6	11.5	83.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	52.3	57.2	62.5	69.0	76.6	82.5	84.7	85.3	79.8	70.9	60.9	53.6
	DBStd	13.73	9.22	10.14	8.81	6.96	5.48	3.75	3.05	3.18	5.15	7.80	8.98	9.37
	HDD50	228	83	44	14	1	0	0	0	0	0	1	18	67
	HDD65	1433	402	252	151	39	3	0	0	0	1	38	177	370
	CDD50	7409	155	245	402	570	825	974	1077	1093	893	648	347	180
	CDD65	3141	9	33	75	159	363	524	612	628	444	220	56	18
	CDH74	32762	57	230	466	1267	3206	5740	7309	7648	4472	1916	358	93
	CDH80	15794	3	52	94	390	1315	2855	3984	4242	2145	656	51	7
Wind	WSAvg	8.6	8.8	9.0	9.6	9.7	9.7	9.2	8.6	7.7	7.0	8.0	8.1	8.3
Precipitation	PrecAvg	33.9	2.0	1.9	2.5	2.5	4.1	3.6	2.8	2.3	3.3	4.0	2.5	2.2
	PrecMax	50.7	5.7	5.6	6.4	7.1	8.5	12.5	14.7	8.1	10.7	20.3	10.6	13.4
	PrecMin	15.3	0.0	0.1	0.0	0.2	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.2
	PrecStd	10.7	1.7	1.6	1.5	1.9	2.6	3.0	3.4	2.3	2.2	4.3	2.6	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	86.3	86.3	92.8	96.7	101.8	102.1	102.3	100.0	92.4	84.6	80.9
		MCWB	61.6	63.0	65.6	69.7	73.6	74.6	73.4	73.8	71.8	71.6	69.4	65.3
	2%	DB	75.1	80.2	82.4	87.9	93.1	98.1	99.8	100.2	96.4	89.6	81.3	76.8
		MCWB	60.8	63.1	65.9	69.0	73.9	74.4	73.6	74.1	73.0	71.9	67.2	64.8
	5%	DB	71.7	76.4	79.3	84.2	90.3	95.3	98.0	98.6	93.3	86.7	78.5	72.6
		MCWB	60.8	62.8	66.3	68.6	73.7	74.5	73.8	74.0	72.6	70.7	66.4	63.7
	10%	DB	68.1	72.3	75.7	81.3	87.5	92.6	95.5	96.5	90.8	83.9	75.1	69.6
		MCWB	59.3	62.0	64.5	67.5	73.4	75.0	74.3	74.2	73.1	70.0	65.7	63.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	70.2	72.3	76.0	78.9	79.7	79.2	79.2	78.7	77.9	74.2	71.5
		MCDB	71.8	75.2	79.1	83.4	87.2	89.3	87.8	87.1	85.0	83.5	78.6	74.7
	2%	WB	66.9	68.4	70.5	73.9	77.3	78.5	78.2	78.2	77.5	76.3	72.1	69.1
		MCDB	70.6	73.1	76.0	81.2	86.1	87.7	87.2	86.9	85.0	82.0	76.8	72.6
	5%	WB	64.4	67.1	69.2	72.5	76.2	77.8	77.4	77.4	76.7	75.0	70.1	67.4
		MCDB	68.4	72.1	74.1	79.3	84.5	86.8	86.6	86.7	84.3	81.0	74.6	71.2
	10%	WB	61.7	65.4	67.8	71.2	75.1	77.0	76.8	76.8	75.9	73.6	68.3	64.2
		MCDB	66.1	70.1	72.9	77.4	83.0	85.8	86.1	86.3	83.6	79.6	73.2	68.0
Mean Daily Temperature Range	5% DB	MDBR	22.3	21.7	21.0	21.9	19.7	20.0	21.5	22.0	21.0	21.9	21.1	21.1
		MCDBR	26.7	25.3	23.0	25.1	22.2	23.5	24.8	24.7	24.2	22.8	23.1	24.5
		MCWBR	15.6	14.1	11.6	11.4	7.3	5.3	5.3	5.3	6.5	9.0	12.5	15.0
	5% WB	MCDBR	20.5	19.1	18.4	18.3	18.4	18.5	20.6	20.9	18.3	17.3	18.7	19.8
MCWBR		14.8	13.1	11.7	9.9	7.7	5.9	5.4	5.3	6.0	9.0	12.4	15.1	
Clear-Sky Solar Irradiance	taub		0.330	0.334	0.360	0.401	0.447	0.429	0.438	0.423	0.415	0.373	0.362	0.318
	taud		2.487	2.482	2.421	2.323	2.238	2.343	2.353	2.375	2.352	2.456	2.417	2.533
	Ebn at Noon		288	296	294	284	270	273	270	274	273	279	273	287
	Edh at Noon		29	32	36	41	45	40	40	38	38	32	31	27
All-Sky Solar Radiation	RadAvg		954	1115	1417	1687	1827	2086	2102	2006	1648	1369	1027	846
	RadStd		106	201	140	165	178	133	181	114	164	180	115	107

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155

Nomenclature: See separate page